Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to improve flood prediction and disaster preparedness using machine learning, enhancing the accuracy and efficiency of flood risk assessment. It addresses limitations in traditional forecasting methods by providing early warnings and supporting effective disaster response. Key features include a machine learning-based flood prediction model and real-time flood risk classification.

Project Overview	
Objective	The primary objective is to develop an intelligent flood prediction system using machine learning techniques to provide accurate flood risk predictions and support timely disaster management decisions.
Scope	The project focuses on analysing historical weather data and developing a machine learning-based flood prediction system integrated with a web application for monitoring flood risks and supporting disaster preparedness.
Problem Statement	
Description	Traditional flood forecasting methods often lack accuracy and timely predictions, making it difficult for authorities to prepare for and respond effectively to flood events.
Impact	Addressing these issues will improve disaster preparedness, reduce the impact of floods, support efficient resource allocation, and help protect lives and property in vulnerable regions.
Proposed Solution	
Approach	Employing machine learning techniques to analyse meteorological data and predict flood occurrence, creating an intelligent and scalable flood prediction system.
Key Features	- Implementation of a machine learning-based flood prediction model.

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	CPU/GPU for model training and testing
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	Storage for datasets, trained models, and logs
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Historical flood and weather datasets, CSV format